

## APPENDIX A. TRAINING AND IMPLEMENTATION DETAILS

## Training loop summary

## Algorithm A1 Physics-aligned MoE training

Input: history windows  $X$ , graph sequence  $A$ , targets  $Y$ , primary regime anchors  $R$ , auxiliary wake labels  $W$  (WTB only), model parameters  $\theta$

for each minibatch do

Build node-level graph context from the precomputed graph tensors

Encode  $X$  with the directed-diffusion GRU to obtain node context  $H_t$

Compute gate logits  $Z$  and soft routing probabilities  $G$

Compute expert forecasts and aggregate them with  $G$

Evaluate  $L_{\text{pred}}$

If MoE mode is active:

add  $L_{\text{bal}}$  on top-K routing statistics

add  $L_{\text{align}}$  on valid primary regime anchors

if WTB: add  $L_{\text{force}}$  on MPPT/pitch samples

if WTB: add  $L_{\text{aux}}$  on wake labels

add  $L_{\text{smooth}}$  on the final graph snapshot

Update  $\theta$  with AdamW and mixed precision

end for

Select the checkpoint with the best validation RMSE

The wake weight is  $\tilde{\mathcal{A}}_t(i, j) = \exp(-d_{ij,t}^\parallel / \alpha) \exp(-|d_{ij,t}^\perp| / \beta) \mathbb{I}_{ij,t}^{\text{cone}}$ . If wind direction is missing, fall back to  $\mathcal{A}^{\text{static}}(i, j) = \exp(-\|\Delta \mathbf{p}_{ij}\|_2 / d_{\text{max}})$ . Only the strongest  $M$  inbound weights are retained. The wake score and flag are  $s_{ij,t}^{\text{wake}} = \sum_j \mathcal{A}_t(i, j)$  and  $\mathbf{r}_{ij,t}^{\text{wake}} = \sum_j \mathbf{r}_{ij,t}^{\text{wake}}$ .

$$W_{i,t} = \begin{cases} 1, & s_{i,t}^{\text{wake}} \geq q_{0.75}^{\text{wake}} \wedge R_{i,t}^{\text{wtb}} \in \{1, 2\}, \\ 0, & s_{i,t}^{\text{wake}} < q_{0.75}^{\text{wake}} \wedge R_{i,t}^{\text{wtb}} \in \{1, 2\}, \\ \emptyset, & \text{otherwise.} \end{cases}$$

For ERA5, the retained symmetric graph uses a Gaussian kernel on great-circle distances:

$$\mathcal{A}(i, j) = \exp\left(-\frac{d_{ij}^2}{2\sigma^2}\right) \mathbf{1}[j \in \mathcal{N}_{k_{\text{nn}}}(i) \text{ or } i \in \mathcal{N}_{k_{\text{nn}}}(j)],$$

where  $\sigma$  is the median retained neighbor distance on the training graph.

## Shared constants

## Model-family and supplementary evidence index

TABLE I  
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| Table A1. In-family model comparison used for mechanism validation. |                      |                                                                                                          |                            |                           |
|---------------------------------------------------------------------|----------------------|----------------------------------------------------------------------------------------------------------|----------------------------|---------------------------|
| Model                                                               | Routing              | Extra terms                                                                                              | Params                     | Role                      |
| Dense (matched)                                                     | Single dense head    | None                                                                                                     | WTB 110,012 / ERA5 103,272 | Shared-mapping reference  |
| Unconstrained MoE                                                   | Node-level soft gate | Prediction loss only                                                                                     | WTB 110,012 / ERA5 103,843 | Routed-capacity control   |
| Corrected routing comparator                                        | Node-level soft gate | WTB boundary-forced: $L_{\text{bal}} + L_{\text{align}} + L_{\text{force}}$ ; ERA5: full corrected stack | WTB 110,012 / ERA5 103,843 | Accountability comparator |

## Auxiliary losses

Let  $\mathcal{B}$  denote the routed samples after masks, with  $B = |\mathcal{B}|$ . The soft importance and normalized share of expert  $e$  are  $I_e = \frac{1}{B} \sum_{n=1}^B g_n^{(e)}$  and  $P_e = I_e / \sum_r I_r$ . The top- $K$  load is  $f_e = \frac{1}{B} \sum_{n=1}^B \mathbf{1}[e \in \text{TopK}(\mathbf{g}_n)]$ , giving

$$\mathcal{L}_{\text{bal}} = E \sum_{e=1}^E f_e P_e - 1.$$

For WTB wake supervision,

$$\mathcal{L}_{\text{aux}} = \frac{1}{|\Omega_{\text{wake}}|} \sum_{(i,t) \in \Omega_{\text{wake}}} \text{BCE}(z_{i,t}^{(\text{wake})}, W_{i,t}),$$

where  $\Omega_{\text{wake}}$  contains only MPPT and pitch-control samples with defined wake flags. The graph-smoothness penalty is

$$\mathcal{L}_{\text{smooth}} = \frac{\sum_{t \in \mathcal{T}_{\mathcal{B}}} \sum_{i,j} \mathcal{A}_t(i, j) \|\mathbf{g}_{i,t} - \mathbf{g}_{j,t}\|_2^2}{\sum_{t \in \mathcal{T}_{\mathcal{B}}} \sum_{i,j} \mathcal{A}_t(i, j) + \epsilon}.$$

## Graph construction details

For WTB, the downstream unit vector is  $\mathbf{u}_{i,t} = [\sin(\theta_{i,t} + \pi), \cos(\theta_{i,t} + \pi)]^\top$ . With  $\Delta \mathbf{p}_{ij} = \mathbf{p}_j - \mathbf{p}_i$ , the streamwise and cross-stream distances are  $d_{ij,t}^\parallel = \Delta \mathbf{p}_{ij}^\top \mathbf{u}_{i,t}$  and  $d_{ij,t}^\perp = |\Delta p_{ij}^x u_{i,t}^y - \Delta p_{ij}^y u_{i,t}^x|$ . A candidate edge activates when

$$\mathbb{I}_{ij,t}^{\text{cone}} = \mathbf{1}\left[d_{ij,t}^\parallel > 0 \wedge \arctan\left(\frac{d_{ij,t}^\perp}{\max(d_{ij,t}^\parallel, 10^{-6})}\right) \leq \phi\right].$$

TABLE II  
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Table A3. Shared architecture, graph, and training constants used in the reported experiments.

| Item                                      | WTB                | ERA5                            | Role                                      |
|-------------------------------------------|--------------------|---------------------------------|-------------------------------------------|
| History / horizon $(H, P)$                | (36, 24)           | (36, 24)                        | Input and forecast lengths                |
| Gate softmax temperature $\tau$           | 0.7                | 0.7                             | Soft routing sharpness                    |
| Top- $K$ in $L_{\text{bal}}$              | 1                  | 1                               | Balancing statistic only                  |
| Hidden size $d$                           | 64                 | 64                              | Shared encoder width                      |
| $k_{\text{nn}}$ candidates                | 8                  | 8                               | Initial neighborhood size                 |
| Maximum candidate radius $d_{\text{max}}$ | 1500 m             | not used                        | WTB proximity fallback / candidate filter |
| Wake-cone half-angle $\phi$               | 25°                | not used                        | WTB downstream activation                 |
| Streamwise decay $\alpha$                 | 1200 m             | not used                        | WTB wake attenuation                      |
| Cross-stream decay $\beta$                | 400 m              | not used                        | WTB wake attenuation                      |
| Retained inbound edges $M$                | 5                  | symmetric $k_{\text{nn}}$ graph | Final graph sparsity                      |
| Optimizer                                 | AdamW              | AdamW                           | Shared optimizer                          |
| Learning rate                             | $2 \times 10^{-3}$ | $2 \times 10^{-3}$              | Shared schedule                           |
| Weight decay                              | $10^{-4}$          | $10^{-4}$                       | Shared regularization                     |
| Batch size                                | 16                 | 16                              | Shared mini-batch size                    |
| Gradient clipping                         | 1.0                | 1.0                             | Shared training stabilization             |

*Dataset-specific thresholds and routing weights*

TABLE III  
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**Table A4.** Dataset-specific regime thresholds used to construct routing anchors.

| Dataset | Threshold                   | Value and meaning                                                            |
|---------|-----------------------------|------------------------------------------------------------------------------|
| WTB     | $u_{\text{idle}}$           | 3.0 m s <sup>-1</sup> , idle / active split                                  |
| WTB     | $u_{\text{rated}}$          | 10.5 m s <sup>-1</sup> , MPPT / pitch boundary                               |
| WTB     | $p_{\text{th}}$             | 2.0 deg, pitch-angle threshold                                               |
| WTB     | $q_{0.75}^{\text{wake}}$    | 1.0690, upper-quartile wake-score threshold                                  |
| ERA5    | $\varepsilon_{\text{sshf}}$ | 169,116, 10th percentile of $ \text{sshf} $ on the training split            |
| ERA5    | $q_{0.95}^{\Delta}$         | 458,287.125, 95th percentile of $ \Delta \text{sshf} $ on the training split |

TABLE IV  
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**Table A5.** Active routing-loss weights in the reported corrected models.

| Weight                    | WTB   | ERA5 | Role                             |
|---------------------------|-------|------|----------------------------------|
| $\lambda_{\text{bal}}$    | 1000  | 0.01 | Expert-usage stabilization       |
| $\lambda_{\text{align}}$  | 5000  | 0.2  | Coarse regime alignment          |
| $\lambda_{\text{force}}$  | 10000 | 0    | Boundary-focused forcing         |
| $\lambda_{\text{aux}}$    | 250   | 0    | Wake-sensitive auxiliary routing |
| $\lambda_{\text{smooth}}$ | 0.05  | 0.05 | Local routing coherence          |